

DEBUGGING ENABLE KARNA / USE KARNA

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INTRODUCTION

DEBUGGING KYA HOTI HAI?

Debugging aik process hai jisme programmer software ya program ke andar mojud **galtiyan (bugs)** dhoondta hai aur unko theek karta hai. Agar koi app ya program sahi result na de, crash ho jaye ya problem create kare to debugging ki madad se us problem ko find kiya jata hai.

BUG KYA HOTA HAI?

Bug ka matlab program ke andar koi **galti ya error** hota hai jo software ko sahi tarah kaam karne se rokta hai.

Example:

Agar calculator mein $5 + 5$ ka answer 20 aaye to ye aik bug hai.



DEBUGGING ENABLE KARNA KYA HAI?

Debugging enable karna ka matlab hai system ya device mein debugging ka option on karna taake developer ko program ki andar ki information mil sake.

Is se developer:

- Errors check kar sakta hai
 - Program ka behavior dekh sakta hai
 - Problems ko solve kar sakta hai
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DEBUGGING KA MAQSAD

Debugging ka main maqsad software ko behtar banana hota hai.

Iske goals:

- Bugs ko remove karna
 - Program ko sahi chalana
 - Performance improve karna
 - Software ko stable banana
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DEBUGGING KA PROCESS__STEP-BY-STEP

1. Problem Dhoondna

Sab se pehle dekha jata hai ke problem kya hai.

Example:

- App open nahi ho rahi
 - Output ghalat aa raha hai
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2. Error Ko Dobara Check Karna

Developer wohi steps dobara karta hai jahan se error aa raha hota hai.

Is se pata chalta hai:

- Error kab aata hai
 - Error kis wajah se aata hai
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3. Problem Ki Wajah Find Karna

Code aur system ko check karke error ki asal wajah dhoondi jati hai.

Iske liye:

- Code check kiya jata hai
 - Logs dekhe jate hain
 - Debugging tools use hote hain
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4. Error Fix Karna

Problem milne ke baad programmer code mein change karke bug ko remove karta hai.

5. Dobara Testing Karna

Fix ke baad program ko dobara run karke check kiya jata hai ke problem solve hui ya nahi.



DEBUGGING KE TOOLS

DEBUGGER

Debugger aik tool hota hai jo programmer ko code ko line by line check karne mein help karta hai.

Is se:

- Code ruk sakta hai
 - Values check ki ja sakti hain
 - Error ki jagah find hoti hai
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BREAK POINT

Breakpoint aik jagah hoti hai jahan programmer program ko rok deta hai.

Iska use:

- Code ka flow check karna
- Variable ki value dekhna

ke liye hota hai.

LOGS

Logs program ki activity ka record hotay hain.

Logs se pata chalta hai:



- Error kahan aaya
- Program crash kyun hua
- System kya kar raha tha

ERRORS KI TYPES

1. SYNTAX ERROR

Jab code likhne mein rule ki mistake ho.

Example:

- Command ghalat likhna

Is wajah se program run nahi hota.

2. RUNTIME ERROR

Jab program chal raha ho aur error aa jaye.

Example:

- Zero se divide karna
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3. LOGICAL ERROR

Program chal jata hai lekin answer ghalat deta hai.

Example:

Calculation ka formula ghalat hona.



DEBUGGING KI AHMIYAT

Debugging important hai kyun ke:

- Software ki quality improve hoti hai
- Errors kam hotay hain
- Program fast aur stable hota hai
- User ko behtar experience milta hai

DEBUGGING KE FAIDE

- Bugs easily find ho jate hain
- Software zyada reliable hota hai
- Development easy hoti hai
- Future problems kam hoti hain

DEBUGGING KE NUKSANAT

- Zyada time lag sakta hai
- Large programs mein error find karna mushkil hota hai
- Complex bugs solve karna difficult hota hai

ASAAN EXAMPLE

Agar mobile app crash ho rahi hai:

1. Developer problem check karega
2. Error message dekhega
3. Code check karega



4. Bug fix karega

5. App dobara test karega

Ye poora process **debugging** kehlata hai.

Pro Tip (Debugging) 💡

"Debugging karte waqt pehle error message aur logs check karo, kyun ke aksar problem ka clue wahi mil jata hai."

NOTES DONE BY: *SYED GHOU*

